

2025/2026

$$1. a) \lim_{x \rightarrow -\infty} \frac{2x^2 + 3}{-x^3 + x} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow -\infty} \frac{\frac{2x^2}{x^3} + \frac{3}{x^3}}{-\frac{x^3}{x^3} + \frac{x}{x^3}}$$

✓ KI : divide

$$= \lim_{x \rightarrow -\infty} \frac{\frac{2}{x} + \frac{3}{x^3}}{-1 + \frac{1}{x^2}}$$

✓ KI : $\frac{n}{-\infty} \rightarrow 0$

$$= 0_{\text{H.}}$$

JI

③

$$b) \lim_{x \rightarrow \infty} \sqrt{x^2 - 1} - \sqrt{x^2 + 1} \quad \infty - \infty$$

✓ KI : conjugate

$$= \lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 1} - \sqrt{x^2 + 1} \right) \times \frac{\left(\sqrt{x^2 - 1} + \sqrt{x^2 + 1} \right)}{\left(\sqrt{x^2 - 1} + \sqrt{x^2 + 1} \right)}$$

$$= \lim_{x \rightarrow \infty} \frac{(x^2 - 1) - (x^2 + 1)}{\sqrt{x^2 - 1} + \sqrt{x^2 + 1}}$$

✓ KI : expand correctly

$$= \lim_{x \rightarrow \infty} \frac{-2}{\sqrt{x^2 - 1} + \sqrt{x^2 + 1}}$$

$$= 0$$

JI

③

$$\begin{aligned}
 c) \quad & \lim_{x \rightarrow 2^+} \frac{10 - 3x - x^2}{|2 - x|} \\
 &= \lim_{x \rightarrow 2^+} \frac{-(x-2)(x+5)}{-(2-x)} \\
 &= \lim_{x \rightarrow 2^+} \frac{-(\cancel{x-2})(x+5)}{\cancel{x-2}} \\
 &= -7 \quad \times
 \end{aligned}$$

✓ B1: $|2-x| = -(2-x)$

✓ K1: factorise

✓ K1 J1

④

$$\begin{aligned}
 2. \quad & \ln(x+y) = e^{xy} \\
 & \text{at } x=0 \Rightarrow \ln(0+y) = e^0 \\
 & \ln(y) = 1 \\
 & y = e
 \end{aligned}$$

✓ K1: find y

$$\Rightarrow \frac{d}{dx} (\ln(x+y)) = \frac{d}{dx} (e^{xy})$$

$$\frac{1}{x+y} \left(1 + \frac{dy}{dx} \right) = e^{xy} \left(1 \cdot y + x \frac{dy}{dx} \right)$$

✓ K1 differentiate

$$\Rightarrow \frac{1}{0+e} \left(1 + \frac{dy}{dx} \right) = e^0 (1 + e + 0)$$

✓ K1: sub in x, y.

$$\frac{1}{e} + \frac{1}{e} \frac{dy}{dx} = 1 + e$$

$$\frac{1}{e} \frac{dy}{dx} = 1 + e - \frac{1}{e}$$

$$\begin{aligned}
 \frac{dy}{dx} &= e \left[1 + e - \frac{1}{e} \right] \\
 &= e + e^2 - 1 \quad \times
 \end{aligned}$$

Either 1:

J1

④

$$b) f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(2^-) = \lim_{x \rightarrow 2^-} \frac{(x^2 - 3) - (2^2 - 3)}{x - 2}$$

✓ BI : formula

$$= \lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 2^-} \frac{(x \cancel{- 2})(x + 2)}{x \cancel{- 2}}$$

J I

$$= 4 \quad \text{X}$$

$$f'(2^+) = \lim_{x \rightarrow 2^+} \frac{(\sqrt{x-1}) - (\sqrt{2-1})}{x - 2}$$

$$= \lim_{x \rightarrow 2^+} \frac{\sqrt{x-1} - 1}{x - 2} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 2^+} \left(\frac{\sqrt{x-1} - 1}{x - 2} \right) \times \left(\frac{\sqrt{x-1} + 1}{\sqrt{x-1} + 1} \right)$$

$$= \lim_{x \rightarrow 2^+} \frac{(x-1) - 1}{(x-2)(\sqrt{x-1} + 1)}$$

$$= \lim_{x \rightarrow 2^+} \frac{x - 2}{(x \cancel{- 2})(\sqrt{x-1} + 1)}$$

$$= \frac{1}{2}$$

J I

∴ since $\lim_{x \rightarrow 2^+} f'(x) \neq \lim_{x \rightarrow 2^-} f'(x)$, hence

✓ BI

$f(x)$ is not differentiable at $x = 2$

X

✓ J I

⑤

$$3. f(x) = \frac{1}{4}x^4 - 2x^2 + 4.$$

$$f'(x) = 0$$

$$x^3 - 4x = 0. \quad \checkmark \text{ B1}$$

$$x(x^2 - 4) = 0.$$

$$x(x-2)(x+2) = 0.$$

critical values:

$$x = 0$$

$$x = 2$$

$$x = -2$$

J1: all correct.

Nature of critical points

$$f''(x) = 3x^2 - 4 \quad \checkmark \text{ K1: } f''(x)$$

$$\text{i) } f''(0) = -4 < 0$$

$\Rightarrow x=0$ is a maximum. $\checkmark$

$$\text{ii) } f''(2) = 3(2)^2 - 4 > 0$$

$\Rightarrow x=2$ is a minimum $\checkmark$

$$\text{iii) } f''(-2) = 3(-2)^2 - 4 > 0.$$

$\Rightarrow x=-2$ is a minimum $\checkmark$

K1: 2 correct

J1: all correct

Inflection value / change of concavity:

$$f''(x) = 0$$

$$\Rightarrow 3x^2 - 4 = 0$$

$$x = \sqrt{\frac{4}{3}}, -\sqrt{\frac{4}{3}}$$

$$\therefore x = \frac{2}{\sqrt{3}}, -\frac{2}{\sqrt{3}} \quad \times$$

Regions

$$\text{i) concave up: } x \in \left(-\infty, -\frac{2}{\sqrt{3}}\right) \cup \left(\frac{2}{\sqrt{3}}, \infty\right) \quad \times$$

$$\text{ii) concave down: } x \in \left(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right) \quad \times$$

J1

both correct.

(6)

SECTION B

1. $z = x + iy$

$$\frac{z+i}{zi+1+i} = 2.$$

$$z+i = 2(zi+1+i)$$

$$z+i = 2zi + 2 + 2i$$

$$(x+iy)+i = 2i(x+iy) + 2 + 2i$$

$$x+iy+i = 2xi + 2yi^2 + 2 + 2i$$

$$x+iy+i = 2xi - 2y + 2 + 2i \quad \checkmark \text{ K1 : } i^2 = -1$$

compare :

$\Rightarrow$ real part :

$$x = -2y + 2 \quad \text{--- (1)}$$

$$x + 2y = 2$$

$\Rightarrow$ Imaginary part :

$$y + 1 = 2x + 2$$

$$y = 2x + 1 \quad \text{--- (2)}$$

$$y - 2x = 1$$

Solve,

$$x = -2(2x+1) + 2$$

$$x = -4x - 2 + 2$$

$$5x = 0$$

$$\therefore x = 0 \quad \checkmark \text{ J1}$$

$$y = 2(0) + 1$$

$$\therefore y = 1 \quad \checkmark \text{ J1}$$

(5)

$$2. \log_4(x+4) - 5 \log_{(x+4)} 16 = 4$$

$$\log_4(x+4) - 5 \left[\frac{\log_4 16}{\log_4(x+4)} \right] = 4 \quad \checkmark B1: \text{change of base}$$

$$\text{Let } R = \log_4(x+4)$$

$$\Rightarrow R - 5 \left(\frac{2}{R} \right) = 4$$

$$R^2 - 10 = 4R$$

$$R^2 - 4R - 10 = 0$$

$$R_1 = 2 + \sqrt{14}$$

$$R_2 = 2 - \sqrt{14}$$

$\checkmark K1$: quadratic eqn

$\checkmark K1$

$$\Rightarrow \log_4(x+4) = 2 + \sqrt{14}$$

$$x+4 = 4^{2+\sqrt{14}}$$

$$\therefore x = 4^{2+\sqrt{14}} - 4$$

$\checkmark J1$

$$\Rightarrow \log_4(x+4) = 2 - \sqrt{14}$$

$$x+4 = 4^{2-\sqrt{14}}$$

$$x = 4^{2-\sqrt{14}} - 4$$

~~XX~~

$\checkmark J1$: must be exact

(5)

$$b) \frac{|x^2-1|}{3x^3-x^2-8x-4} \leq 0$$

$$\frac{|(x-1)(x+1)|}{(x-2)(x+\frac{2}{3})(x+1)} \leq 0$$

✓ KI : factorise

Roots $x=1, -1$

$x=2, -\frac{2}{3}, -1$

✓ KI : correct at least 3 roots.

	-1	$-\frac{2}{3}$	1	2	
$ x-1 $	+	+	+	0	+
$ x+1 $	+	0	+	+	+
$x-2$	-	-	-	-	0
$x+\frac{2}{3}$	-	-	0	+	+
$x+1$	-	0	+	+	+
	-	+	-	-	+

✓ KI : method

KI

⑤

$$\therefore x \in (-\infty, -1] \cup (-\frac{2}{3}, 2) \quad \text{JI}$$

- 3 a) (i) many-to-one ✓ BI
- (ii) one-to-one ✓ BI

②

$$b) (f \circ g)(x)$$

$$= f(g(x))$$

$$= f(2^{x-3})$$

$$= (2^{x-3})^3$$

$$= 2^{3x-9}$$

✓ KI : composite

✓ JI

Hence, $(g^{-1} \circ f^{-1})(x) = (f \circ g)^{-1}(x)$

✓ B1 : composite properties

Find $(f \circ g)^{-1}(x)$

* If B0 here $\rightarrow$ J0

$\Rightarrow$ Let $h(x) = (f \circ g)(x)$

$$h(x) = 2^{3x-9}$$

$$h(h^{-1}(x)) = x$$

$$2^{3h^{-1}(x)-9} = x$$

✓ B1 : inverse definition

$$\ln 2^{(3h^{-1}(x)-9)} = \ln x$$

$$(3h^{-1}(x)-9) \ln 2 = \ln x$$

✓ K1

$$3h^{-1}(x)-9 = \frac{\ln x}{\ln 2}$$

$$\therefore h^{-1}(x) = \frac{1}{3} \left(\frac{\ln x}{\ln 2} + 9 \right) \text{ or } \frac{\ln x}{3 \ln 2} + 3 \quad \times$$

Alternatively,

$$2^{3h^{-1}(x)-9} = x$$

$$\log_2 2^{3h^{-1}(x)-9} = \log_2 x$$

$$3h^{-1}(x)-9 = \log_2 x$$

$$h^{-1}(x) = \frac{1}{3} \log_2 x + 3 \quad \times$$

Hence,

$$(g^{-1} \circ f^{-1})(x) = (f \circ g)^{-1}(x) = h^{-1}(x)$$

$$= \frac{\ln x}{3 \ln 2} + 3 \quad \times$$

J1

⑥

4. $m(x) = 3e^{-x} - 4$

a) domain $x \in (-\infty, \infty)$ J1
 range $y \in (-4, \infty)$ J1

②

b) $m^{-1}(x)$ exists if $m(x)$ is a 1-to-1 function.

$$m(a) = m(b)$$

$$3e^{-a} - 4 = 3e^{-b} - 4$$

$$3e^{-a} = 3e^{-b}$$

$$e^{-a} = e^{-b}$$

$$\Rightarrow -a = -b$$

$$\therefore a = b$$

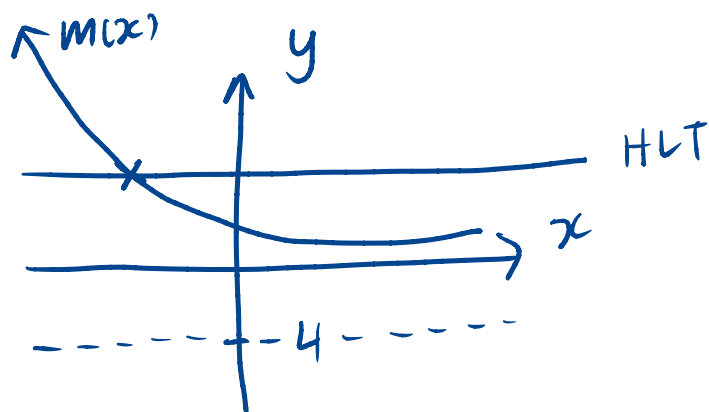
K1

J1

Since $m(a) = m(b)$ gives $a = b$,
 thus $m(x)$ is a 1-to-1 function.
 Hence $m^{-1}(x)$ exists.

} BI : both

Alternatively,



by using Horizontal Line Test, $m(x)$ only
 intersect once with the line.

Hence $m(x)$ is 1-to-1.

Thus, $m^{-1}(x)$ exists.

③

$$c) m(m^{-1}(x)) = x.$$

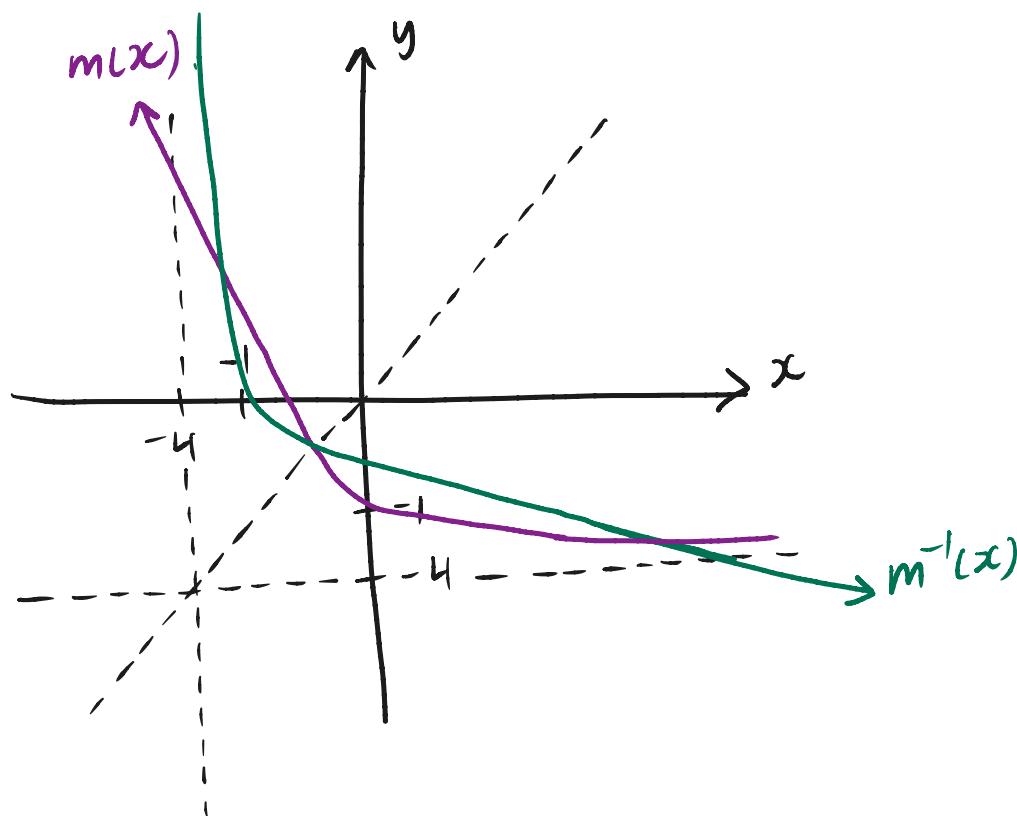
$$3e^{-m^{-1}(x)} - 4 = x$$

✓ B1 : inverse definition

$$e^{-m^{-1}(x)} = \frac{1}{3}(x+4)$$

$$\ln e^{-m^{-1}(x)} = \ln\left(\frac{x+4}{3}\right) \quad K1$$

$$m^{-1}(x) = -\ln\left(\frac{x+4}{3}\right) \quad J1$$



R1 : $m(x)$ shape

R1 : $m^{-1}(x)$ shape

R1 : $\begin{cases} 1. \text{ correct } x, y\text{-asymptotes,} \\ 2. \text{ intersect at axis of symmetry} \\ 3. \text{ complete labels.} \end{cases}$

⑥

$$5 a) f(x) = \frac{x^2 - 4}{q - x}$$

(i) vertical asymptote
 $x = q$ ✓ J |

$$\rightarrow \begin{array}{c} | \\ \hline q \end{array} \leftarrow$$

$$\lim_{x \rightarrow q^-} \frac{x^2 - 4}{q - x} = +\infty$$

$$\lim_{x \rightarrow q^+} \frac{x^2 - 4}{q - x} = -\infty$$

✓ B | : $\lim_{x \rightarrow q^\pm} f(x)$

(ii) Horizontal asymptote
 $y = ?$

$$y = \lim_{x \rightarrow \infty} \frac{x^2 - 4}{q - x}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x} - \frac{4}{x}}{\frac{q}{x} - \frac{x}{x}}$$

✓ K | : $\lim_{x \rightarrow \pm\infty} f(x)$

$$= \lim_{x \rightarrow \infty} \frac{x - \frac{4}{x}}{\frac{q}{x} - 1}$$

$\rightarrow -\infty$ (undefined) ✓ J |

✓ B |

Hence, horizontal asymptote does not exist.

Thus $f(x)$ is continuous $\forall x \in (-\infty, \infty) / \{q\}$ ✓ J |

similarly : $x \in (-\infty, q) \cup (q, \infty)$

⑥

b)

$$\begin{array}{c} x^2 - 4x + 3k \quad 2e^x - 5k \quad \frac{2x-1}{x+2} \\ \hline 0 \qquad \qquad \qquad 3 \end{array}$$

(i) $\lim_{x \rightarrow 0^-} h(x) = \lim_{x \rightarrow 0^-} x^2 - 4x + 3k \quad \checkmark K1$
 $= 3k \quad \checkmark J1$

$\lim_{x \rightarrow 0^+} h(x) = \lim_{x \rightarrow 0^+} 2e^x - 5k$
 $= 2(1) - 5k$
 $= 2 - 5k \quad \checkmark J1$

Since $h(x)$ is continuous $\Rightarrow \lim_{x \rightarrow a^-} f(x) = f(a)$

$\lim_{x \rightarrow 0^-} h(x) = \lim_{x \rightarrow 0^+} h(x) \quad \checkmark B1$

$3k = 2 - 5k$

$8k = 2$

$k = \frac{1}{4} \quad \checkmark J1$

⑤

(ii) $f(3) = \frac{2(3)-1}{(3)+2}$
 $= \frac{5}{5}$
 $= 1$

$\checkmark K1$: find $f(3)$

$\lim_{x \rightarrow 3^-} f(x) \neq 1 \quad \checkmark B1$

$\therefore$ Since $\lim_{x \rightarrow 3^-} f(x) \neq f(3)$, hence $f(x)$ is not continuous at $x=3$ $\checkmark$

$\checkmark J1$

③

$$6, \quad x = \frac{1}{\sqrt{1+t^2}}$$

$$a)(i) \quad x = (1+t^2)^{-1/2}$$

$$\frac{dx}{dt} = -\frac{1}{2} (1+t^2)^{-3/2} (2t)$$

$$= \frac{-t}{(1+t^2)^{3/2}} \quad \checkmark \text{ KI!}$$

$$y = \frac{\sqrt{1+t^2}}{t}$$

$$= \underbrace{(1+t^2)^{1/2}}_u \cdot \underbrace{t^{-1}}_v$$

$$u = (1+t^2)^{1/2} \quad v = t^{-1}$$

$$u' = \frac{1}{2} (1+t^2)^{-1/2} (2t) \quad v' = -t^{-2}$$

$$= \frac{t}{\sqrt{1+t^2}} \quad \checkmark \text{ KI! correct } u' \text{ or } v'$$

$$\frac{dy}{dt} = \frac{t}{\sqrt{1+t^2}} (t^{-1}) + \sqrt{1+t^2} (-t^{-2})$$

$\checkmark \text{ KI! formula}$

$$= \frac{1}{\sqrt{1+t^2}} - \frac{\sqrt{1+t^2}}{t^2}$$

$$= \frac{t^2 - (\sqrt{1+t^2})^2}{t^2 \sqrt{1+t^2}}$$

$$= \frac{-1}{t^2 \sqrt{1+t^2}} \quad \checkmark \text{ JI}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \frac{-1}{t^2 \sqrt{1+t^2}} \times \left(-\frac{(1+t^2)^{3/2}}{t} \right) \quad \checkmark \text{ KI}$$

$$= \frac{1+t^2}{t^3} \quad \checkmark \text{ BI}$$

⑥

(ii) $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$ correct u' or v'
 $= \frac{d}{dt} \left(\frac{1+t^2}{t^3} \right) \frac{dt}{dx}$ ✓ BI u = 1+t^2
 $= \left(\frac{u'v - uv'}{v^2} \right) \frac{dx}{dt}$ u' = 2t ✓ KI
 $= \frac{2t(t^3) - (1+t^2)(3t^2)}{(t^3)^2} \times \left(-\frac{(1+t^2)^{3/2}}{t} \right)$ v = t^3 v' = 3t^2 correct u' & v'
✓ JI: both u' & v' ✓ KI: formula

when $t=1$

$\frac{d^2y}{dx^2} = \left(\frac{2(1) - (2)(3)}{(1)} \right) \times -\frac{(2)^{3/2}}{1}$ ✓ KI: t=1
 $= 8\sqrt{2} \times$ ✓ JI ⑥

b) $y = \sin(x^2+1)$

$\frac{dy}{dx} = \cos(x^2+1) (2x)$ ✓ JI: dy/dx
 $= \frac{2x \cos(x^2+1)}{u}$ v

$\frac{d^2y}{dx^2} = (2) \cos(x^2+1) + 2x(-\sin(x^2+1) 2x)$ ✓ KI: product rule
 $= 2 \cos(x^2+1) - 4x^2 \sin(x^2+1)$ ✓ JI

$\Rightarrow \left(\frac{d^2y}{dx^2} = \frac{1}{x} \frac{dy}{dx} - 4x^2 y \right) x$ ✓ KI: substitute

$x \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3 y = 0$

✓ BI: proven ⑤

7 (a) $f(x) = (x-2)(3+2x)^2$

(i) y-intercept: $x=0$

$$y = (-2)(3)^2$$

$$y = -18 \quad \checkmark \text{ J1}$$

①

$$\begin{aligned} \text{(ii)} \quad f'(x) &= (1)(3+2x)^2 + (x-2)(2)(3+2x)'(2) \\ &= (3+2x)^2 + 4(x-2)(3+2x) \quad \checkmark \text{ K1} \\ &= (3+2x)(3+2x+4(x-2)) \\ &= (3+2x)(-5+6x) \quad \checkmark \text{ J1} \end{aligned}$$

Hence,

$$f'(x) = 0$$

$$\Rightarrow (3+2x)(-5+6x) = 0$$

$$x = -\frac{3}{2}, \quad x = \frac{5}{6} \quad (\text{zero gradient})$$

x	$(-\infty, -\frac{3}{2})$	$-\frac{3}{2}$	$(-\frac{3}{2}, \frac{5}{6})$	$\frac{5}{6}$	$(\frac{5}{6}, \infty)$	$\checkmark \text{ K1}$
$f'(x)$	+	0	-	0	+	check gradient

Thus,

f increases: $x \in (-\infty, -\frac{3}{2}) \cup (\frac{5}{6}, \infty)$ $\checkmark \text{ J1}$

f decreases: $x \in (-\frac{3}{2}, \frac{5}{6})$ $\checkmark \text{ J1}$

⑤

b) (i) Info:

$$0 \leq h \leq 25$$

$$V = 4\pi r h (h+4)$$

$$\frac{dV}{dt} = 80\pi$$

$$\text{Goal: } \frac{dh}{dt} = ? \text{ at } h=6.$$

(ii) Eqn: ?

$$\frac{dh}{dt} = \boxed{\frac{dh}{dV}} \times \frac{dV}{dt} \quad \checkmark \text{BI}$$

Recall:

$$\begin{aligned} V &= 4\pi r h (h+4) \\ &= 4\pi r h^2 + 16\pi r h \end{aligned}$$

$$\Rightarrow \frac{dV}{dh} = 8\pi r h + 16\pi r \quad \checkmark \text{KI}$$

Hence,

$$\frac{dh}{dt} = \frac{1}{8\pi r h + 16\pi r} \times (80\pi) \quad \checkmark \text{KI}$$

$$= \frac{1}{64} (80) \quad \checkmark \text{KI}$$

$$= 1.25 \text{ cm s}^{-1} \quad \checkmark \text{JI}$$

⑤